B5: E-Graph-Augmented Tool-in-the-Loop RTL Verification

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P2: Prototype & Pilot Results | Due November 9, 2026, 11:59 PM ET

I. Project Goal and Updates

Our project asks whether E-graph semantic feedback helps an LLM verify RTL more effectively and efficiently than raw execution feedback. The proposal compares text-only (A), raw-coverage (B), and E-graph-enhanced (C) stimulus generation. Exploratory coverage pilots implement that original track. The separately authorized *test2* extension fixes stimuli and generates passive SystemVerilog monitors, measuring valid fault detection, false alarms, and cost. This separates a generated checker's quality from the model's input-selection ability while testing the same semantic-feedback hypothesis.

The endpoint change is explicit: changing a monitor under fixed stimuli does not improve DUT coverage. Raw RTL counters provide workload context; monitor quality is the outcome. We do not claim an official CVDP or Pass@1 evaluation. For P2, we now have a completed, independently audited controller retest of the implemented pipeline. It reuses a published benchmark, so it is not a new hidden test or cross-IP generalization result. The current scope ends at this milestone; cross-IP execution remains paused.

II. Method Overview

We freeze the DUT, specification, input traces, independent reference, model settings, selection rule, and resource limits before comparing B and C. Python orchestrates candidate generation, compilation, public development feedback, per-round snapshots, release, and common grading. A Rust `egg` pipeline forms restricted equivalence groups; independent Z3 checks preserve widths, signedness, complete activation guards, and clock/reset context. These establish limited two-state combinational relations, not general sequential or four-state equivalence. Equivalent guards retain their original source counters.

A Claude-authored specification checker remains independent of GPT candidate monitors; candidates never grade themselves. B receives the permitted specification/RTL, fixed development observations, raw counters, and its own previous candidate/development feedback. C adds validated E-graph relations. The current prompts exclude reference code, mutant sources, final evaluation observations, and solver witnesses. Withholding them from these requests does not make the previously published corpus fresh hidden data.

The new retest uses `gpt-6.1-sol`/low, one independent session per arm and three rounds, interleaved B1,C1,C2,B2,B3,C3. Model and registered resource policy are shared, including an 8,192-output-token ceiling; equal limits do not mean equal realized tokens. All six candidates freeze before one grading pass, and each final is its last compile-valid candidate, never selected by final score. A final-post `trace_done` marker exposes finite trace termination without adding a DUT input or clock edge.

Correct-DUT controls gate fault credit. A credited alarm must align with an independent observable violation; an earlier alarm cannot earn credit. Unknowns, invalid candidates, unobservable faults, and failures remain in the evidence. Repeated alarms and two simulators do not create extra defect types. Tokens, candidate compilation, DUT simulation, monitor replay, and wall time are distinct measurements. Without a token-matched placebo, B/C tests the whole feedback package, not pure semantic content.

III. Prototype and Pilot Results

Current end-to-end prototype. The multi-cycle request/response controller uses frozen inputs under `test2/tasks/axis_length_done_v1/`. The new registration is `test2/configs/controller_done_bc_retest_v3.json`; the entry is `test2/retest_v3/cli.py`. The saved run in `test2/results/controller-done-bc-retest-v3/` binds prompts, responses, `release.json`, final selections, usage, and scores. It completed once: six generations, compilation/development feedback, all-round freeze, then common grading and report. There were no retries or remote token-count calls. P19 independently checked saved evidence; it did not generate another cohort.

Earlier infrastructure failures remain archived. The historical pilot's catalog-field audit failure required read-only recovery; a later adapter qualification rejected missing returned service tiers. Those versions and counterexamples are preserved. The accepted R03 adapter repaired that gate before the current live retest; no old outcome was relabeled.

Latest comparison. Both arms already reach the same case/type union in round 1 and retain it in rounds 2–3 (Fig. 1); both finals select round 3. Each candidate passes 18 finite correct controls. The raw inventory is 12 fault types/144 fault pairs plus 18 controls, or 162 cases per candidate. All 38 reference-unknown pairs remain per round. N10 is a mutant identifier: its 12 frozen reference outcomes are PASS, with no eligible fault evidence. Separately, input completion is 0/10 eligible N01 pairs, all retaining reference unknown. Neither fact is silently converted into a pass or removed from the raw

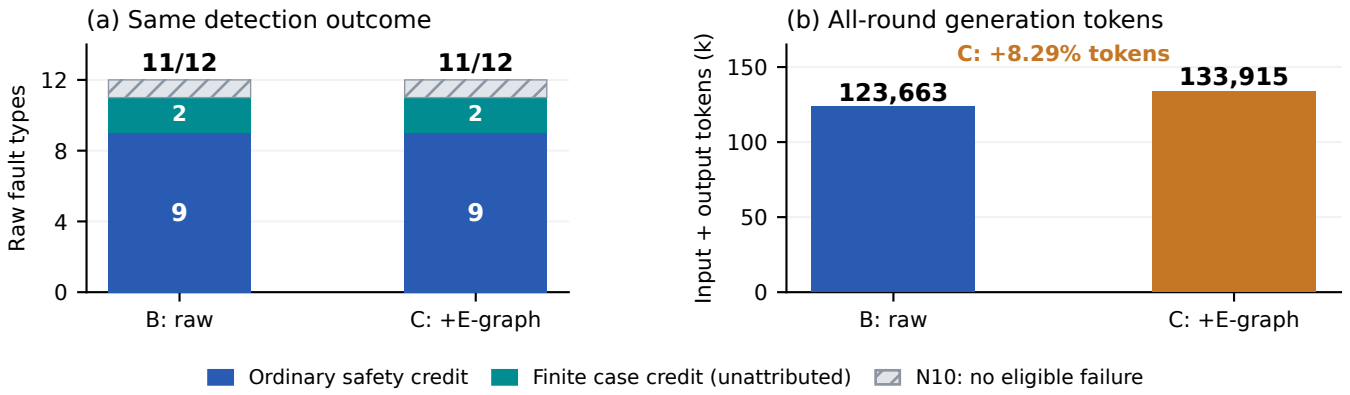

Fig. 1. P19-audited controller retest: one independent session per arm, three rounds each (six generation calls), on the same published fixed-stimulus corpus. Left: both final monitors have the same raw union of 11/12 fault types: nine ordinary-safety types plus two end-marker finite case types, *unattributed* to a specific monitor property. Gray N10 has no eligible reference failure and remains in the raw denominator. Right: actual generation input+output totals over all three rounds, B 123,663 and C 133,915 (+8.2903%). No averaging or independent-repeat error bars are implied. Cache/reasoning tokens are subsets, not added again; this is not an invoice or a correctness proof.

denominator. The finite credit is case-level and unattributed; passing controls does not prove general monitor soundness or 128-cycle completion.

The new run uses B 116,529 input + 7,134 output tokens and C 127,168 + 6,747. C spends 8.2903% more total tokens without extra detections. The combined conservative usage-based cost upper is $1.569371, not an actual provider invoice. The run took 177.995 s including cleanup, with 12 candidate compiler processes and 990 monitor replays (18 development + 972 grading), covering 204,444 replayed cycles. These reuse frozen DUT observations: they are not new physical DUT simulations, generated stimuli, or formal proofs.

Historical cohort, kept separate. The earlier three-sessions-per-arm, six-round pilot used 36 generation calls. B totaled 740,643 tokens and C 813,649 (+9.8571%), with the same final union of 11 observable types and raw denominator 12. Its invalid rounds and costs remain; one B final has 40 total unresolved pairs rather than 38. The earlier 540 formal jobs still have 75 PASS, 104 FAIL, and 361 TIMEOUT statuses, including 74 unreproduced/unresolved BMC witnesses. The new retest does not resolve those outcomes. Neither cohort was rerun to prepare this paper, and the cohorts are not pooled as independent fresh-hidden evidence.

The editable source includes `figure-data.json`, which binds the retest ledger/grade, historical audit, and P19 review by hash. All figure values derive from these saved artifacts; no research experiment was rerun for this paper.

IV. Next Steps

Main risk and paper plan. Saturation leaves little room for C to improve; one session per arm, one published workload, and longer C context limit causal and statistical conclusions. This run shows no additional C benefit here, not that E-graphs are universally unhelpful. G1 now has P17's limited finite-control acceptance; that is not a proof of these monitors or a reason to repeat generic qualification for P2.

A possible Milestone 3 plan (Nov 23), subject to separate future authorization, would preregister a small nonsaturated obligation set and a genuinely disjoint holdout before seeing C outcomes, then compare matched conditions and a token-matched placebo. Any new oracle scope must receive independent Claude review; GPT will not silently repair it. Existing cross-IP task drafts are preparation, not implemented or tested multi-IP monitor results. No new experiment or paid call is promised by this P2 delivery.

Actual midterm feedback/response pending. Authentic instructor/TA technical feedback has not been found in the bounded local review. We cannot invent a response: the original comment, resulting change, evidence, and limit must be recorded when available. User priorities and internal P19 audits do not substitute for classroom feedback. The two-page draft therefore retains this content gap and has not been submitted to the course platform.